From Predictive Uncertainty to Decision-Effective Dynamics Ambiguity Sets

Robust Control of Function-Valued Systems under Deployment Shift

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Abstract

Learned dynamics can be calibrated in a predictive norm yet still induce poor control when errors align with sensitive cost-to-go directions. We ask how predictive uncertainty can become a decision-effective dynamics ambiguity set. A base world model and a label-perturbed replica output a mean and a smoothed disagreement scale; a small deployment audit conformalizes the resulting anisotropic ellipsoid or simultaneous coordinate box. Projected gradient ascent over this set gives a nonlinear adversarial controller, while its closed-form support in the finite-horizon adjoint direction gives a fast first-order approximation. We establish marginal one-step coverage, derive the exact support and linearization remainder, and state rollout and discounted value-error bounds under Lipschitz assumptions. A sharp scalar witness verifies linear scaling in one-step error and quadratic scaling in the effective discount horizon. Controlled viscous Burgers dynamics test viscosity, boundary, and latent actuator-gain shifts; the official two-dimensional Navier–Stokes benchmark is reserved for function-valued uncertainty because it has no control channel. The current experiments are mechanism evidence, not a population-level control claim.

1 Introduction

Fast learned surrogates make repeated planning for distributed-parameter systems increasingly plausible. Neural operators, for example, learn function-to-function maps and can replace expensive numerical time steppers in model predictive control (MPC) [6]. Their deployment creates a separate difficulty: a controller optimizes against a learned model and can therefore exploit precisely those errors that appear harmless under an average prediction metric.

Uncertainty quantification does not resolve this problem by itself. Calibration is a statement about an event and a score; it does not state that the covered directions matter for a decision. This distinction is particularly sharp for a function-valued state. A large error far from a controlled region can have little effect on the objective, whereas a smaller error aligned with a cost-to-go gradient can reverse the preferred action. Related work has shown that calibrated uncertainty improves model-based planning [7], that model losses can be made value-aware [4], and that uncertainty estimates for PDE surrogates can degrade under moderate distribution shift [8]. These facts motivate a narrower question:

> *How can predictive uncertainty be converted into a dynamics ambiguity set that is effective for control decisions?*

Conformal regions have already been combined with controller robustification [2], robust optimization [5], ensemble-based OOD adaptation [3], and OOD system-level synthesis [10]. We therefore do not claim that conformal robust control is new. Our target is the geometry connecting a calibrated function-space dynamics set to the downstream finite-horizon objective. Unlike the online weighted-conformal constraint-tightening pipeline of [2], the present prototype uses an offline deployment audit, anisotropic function-space sets, and either adjoint support or nonlinear adversarial rollouts. We therefore do not inherit their time-series guarantee; our one-step and behavior-policy trajectory statements are proved separately under exchangeability.

The proposed construction begins with a heteroscedastic predictive ellipsoid, calibrated on a small deployment audit. Robust MPC then solves an approximate inner maximization over a rectangular product of these one-step sets. Automatic differentiation also computes the adjoint of the remaining nominal cost, giving a closed-form first-order alternative. This separates three effects that are often conflated: using deployment data, obtaining predictive coverage, and improving the resulting decision.

Our intended contributions are:

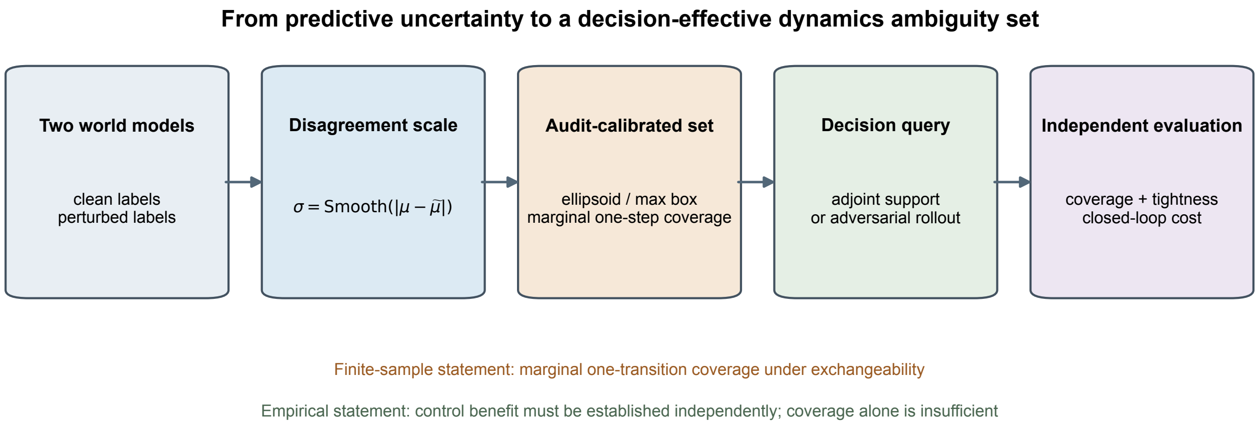

Figure 1: Method and claim hierarchy. A proper-training disagreement field is conformalized on a disjoint deployment audit, queried in decision-sensitive directions, and evaluated for coverage, tightness, and closed-loop cost on independent data. Coverage and control benefit are separate statements.

1. an audit-calibrated, anisotropic dynamics ambiguity set for function-valued transitions;
2. a nonlinear adversarial-rollout controller and an exact adjoint support formula for its first-order approximation;
3. a transparent hierarchy of finite-sample, deterministic, and first-order theoretical statements; and
4. an evaluation protocol that requires closed-loop benefit beyond coverage, using matched in-distribution, audit-L2, and adjoint-support baselines.

The initial implementation uses a residual Fourier neural operator (FNO) as a world model. The paper is not an FNO architecture contribution.

2 Problem formulation

Let $x_t \in \mathbb{R}^n$ be a discretized function and $a_t \in \mathcal{A}$ a control. The unknown deployment dynamics are

$$x_{t+1} = G_\star(x_t, a_t; \xi),$$

where ξ includes physical parameters, boundary conditions, and latent actuator properties. A learned model is trained on a source distribution, then used to minimize the finite-horizon cost

$$J_H(x_0, a_{0:H-1}) = \sum_{t=1}^{H} \ell(x_t) + \sum_{t=0}^{H-1} r(a_t).$$

At deployment, we assume access to a small independent audit sample $\mathcal{D}_{\text{aud}} = \{(x_i, a_i, y_i)\}_{i=1}^m$, drawn from the deployment regime but not used to update the mean dynamics model.

We call an ambiguity set *decision-effective* if, at a matched audit budget and nominal model, incorporating it into the controller improves a predeclared downstream metric: paired mean closed-loop cost, upper-tail cost, or constraint violation. Predictive coverage is necessary evidence, but is not the definition.

3 From predictive uncertainty to a dynamics ambiguity set

3.1 Function-valued mean and scale

The world model outputs

$$\widehat{G}_\theta(x, a) = \mu_\theta(x, a), \qquad \sigma_\theta(x, a) \in \mathbb{R}^n_{>0}.$$

Our prototype uses residual operator blocks and predicts $\mu_\theta(x,a) = x + \delta_\theta(x,a)$. A second operator has the same architecture and proper-training inputs but is trained once on labels perturbed by fixed Gaussian noise. If its mean is $\widetilde{\mu}_\theta$, the raw scale is a spatially smoothed disagreement

$$\sigma_\theta(x,a) = \mathsf{Smooth}\big(|\mu_\theta(x,a) - \widetilde{\mu}_\theta(x,a)|\big) \vee \sigma_{\min}.$$

This scale is not itself a confidence interval. It only localizes where the world model may be inaccurate; the audit quantile supplies calibration. The proper-training, validation, audit, and test samples are kept disjoint.

3.2 Deployment-audit conformalization

Write $\|v\|_{2,n}^2 = n^{-1}\sum_{j=1}^n v_j^2$. Define the audit score

$$S_i = \left\| \frac{y_i - \mu_\theta(x_i, a_i)}{\sigma_\theta(x_i, a_i)} \right\|_{2,n}.$$

For target miscoverage α, let q be the empirical order statistic with rank $\lceil (m+1)(1-\alpha) \rceil$ in the audit scores augmented by an atom at $+\infty$. Thus $q = +\infty$ if the audit sample is too small for the requested finite-sample level; all experiments below use ranks within the observed audit scores. The ambiguity set is

$$\mathcal{U}_\theta(x,a) = \Big\{ \mu_\theta(x,a) + \sigma_\theta(x,a) \odot z : \|z\|_{2,n} \le q \Big\}. \tag{1}$$

For simultaneous coordinate coverage we additionally use

$$S_i^{\max} = \max_j \frac{|y_{ij} - \mu_{\theta,j}(x_i, a_i)|}{\sigma_{\theta,j}(x_i, a_i)}, \qquad \mathcal{U}_\infty(x,a) = \{\mu_\theta + \Delta : |\Delta_j| \le q_{\max}\sigma_{\theta,j}\ \forall j\}.$$

The corresponding conformal event covers the full discretized output field for one random transition. It still does not imply simultaneous trajectory coverage.

Proposition 1 (Marginal one-step coverage)**.** *Conditional on the trained model, if the audit examples and a new deployment example are exchangeable, then*

$$\Pr\{G_\star(x,a) \in \mathcal{U}_\theta(x,a)\} \ge 1 - \alpha.$$

Proof. The m audit scores and the new score are exchangeable. The rank of the new score is therefore uniform up to ties; the conformal order statistic gives the stated lower bound. □

For a fixed horizon H and a predeclared rollout procedure, define the trajectory score

$$S_i^{\mathrm{traj}} = \max_{0 \le t < H} \max_{1 \le j \le n} \frac{|x_{i,t+1,j} - \widehat{x}_{i,t+1,j}|}{\sigma_{\theta,j}(\widehat{x}_{i,t}, a_{i,t})}.$$

Calibrating this score on independent deployment trajectories gives a distinct trajectory band, rather than pretending that repeated one-step marginal coverage is simultaneous coverage.

Proposition 2 (Marginal trajectory coverage)**.** *Conditional on the trained world model and rollout procedure, if the audit trajectories and a new length-H deployment trajectory are exchangeable, then the split-conformal order statistic of S_i^{traj} covers all times and coordinates of the new rollout with probability at least $1-\alpha$.*

The proposition is distribution-specific: in our experiment the actions come from the behavior policy used to generate the trajectory split. It therefore does not certify counterfactual action sequences selected by MPC.

4 Decision-effective robust MPC

4.1 Nonlinear adversarial rollout

For a candidate action sequence, the rectangular robust planning problem is

$$\min_{a_{0:H-1}} \max_{\|z_t\|_{2,n} \le q} \sum_{t=1}^{H} \ell(x_t) + \sum_{t=0}^{H-1} r(a_t), \quad x_{t+1} = \mu_\theta(x_t, a_t) + \sigma_\theta(x_t, a_t) \odot z_t. \tag{2}$$

The prototype uses projected gradient ascent for $z_{0:H-1}$ and the cross-entropy method for the outer action sequence. This computes an approximate saddle point; finite PGD iterations are not an optimality certificate. The rectangular product is a robust planning choice. Marginal one-step conformal coverage does not imply simultaneous coverage of every set along a trajectory.

4.2 Fast adjoint approximation

Use $\langle v, w \rangle_n = n^{-1} \sum_j v_j w_j$. For a candidate nominal rollout, let

$$\lambda_{t+1} = \nabla_{x_{t+1}} \widehat{J}_H$$

be the full finite-horizon adjoint, including the dependence of later predicted states on x_{t+1}.

Proposition 3 (Exact ellipsoid support)**.** *For every $\lambda \in \mathbb{R}^n$,*

$$\sup_{\Delta: \|\Delta/\sigma\|_{2,n} \leq q} \langle \lambda, \Delta \rangle_n = q \, \|\lambda \odot \sigma\|_{2,n} \,. \tag{3}$$

Proof. Set $\Delta = \sigma \odot z$. Cauchy–Schwarz gives $\langle \lambda \odot \sigma, z \rangle_n \leq \|\lambda \odot \sigma\|_{2,n} \|z\|_{2,n}$. Equality is attained by choosing z parallel to $\lambda \odot \sigma$. □

Linearizing the inner problem in Equation (2) gives

$$\widehat{J}_H(a_{0:H-1}) + \sum_{t=0}^{H-1} q \, \|\lambda_{t+1} \odot \sigma_\theta(\widehat{x}_t, a_t)\|_{2,n} \,. \tag{4}$$

This adjoint-support controller is retained as a computationally cheap approximation and ablation. The nonlinear adversarial rollout is the primary robust controller.

4.3 Linearization remainder

Proposition 4 (Conditional curvature bound)**.** *Suppose the Hessian of the remaining horizon cost with respect to x_{t+1} has operator norm at most M_H throughout $\mathcal{U}_\theta$. Then*

$$\sup_{\Delta \in \mathcal{U}_\theta} J_H(\widehat{x} + \Delta) \leq J_H(\widehat{x}) + q \, \|\lambda \odot \sigma\|_{2,n} + \frac{M_H}{2} q^2 \|\sigma\|_\infty^2.$$

The adjoint controller uses the first-order terms in Equation (4); the adversarial controller instead optimizes the nonlinear rollout numerically. Neither is trajectory-certified: the former omits a certified curvature buffer, while the latter uses a finite-step nonconvex inner solver.

5 Rollout and value-error bounds

Assumption 1. *On a forward-invariant set, $G_\star$ is L_G-Lipschitz in state and $\left\| G_\star(x, a) - \widehat{G}_\theta(x, a) \right\|_{2,n} \leq \epsilon$ uniformly.*

For a common open-loop action sequence and $e_t = \|x_t - \widehat{x}_t\|_{2,n}$,

$$e_{t+1} \leq L_G e_t + \epsilon, \qquad e_h \leq \epsilon \sum_{j=0}^{h-1} L_G^j.$$

If the state-dependent stage reward is L_r-Lipschitz and $\gamma L_G < 1$, then, for the common open-loop action sequence,

$$|V_{G_\star} - V_{\widehat{G}}| \leq \frac{\gamma L_r \epsilon}{(1-\gamma)(1-\gamma L_G)}.$$

Proposition 5 (Policy-transfer bound)**.** *Suppose every policy in the comparison class induces true and learned closed-loop maps sharing the same state Lipschitz bound L_G, the uniform one-step error assumption holds over that class, and the induced closed-loop stage reward $r_\pi(x)$ is uniformly L_r-Lipschitz. If $\widehat{\pi}$ maximizes value in the learned model, then*

$$V_{G_\star}^{\pi^\star} - V_{G_\star}^{\widehat{\pi}} \leq \frac{2\gamma L_r \epsilon}{(1-\gamma)(1-\gamma L_G)}.$$

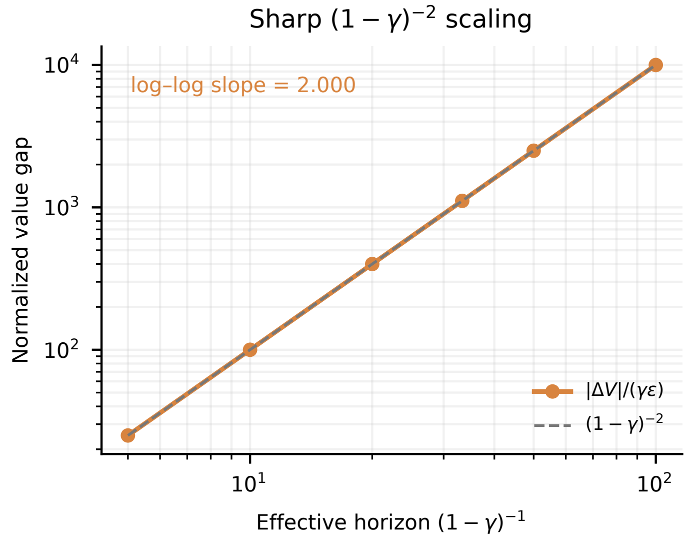

Figure 2: A sharp fixed-policy witness attains the $O(\epsilon/(1-\gamma)^2)$ rate. The plotted value gap is normalized by $\gamma\epsilon$, making the predicted slope exactly two. This is a theorem sharpness check, not a learned-model performance result.

Proof. Let $B = \gamma L_r \epsilon / \{(1-\gamma)(1-\gamma L_G)\}$. The fixed-policy argument above holds uniformly, so

$$\begin{aligned} V_{G_\star}^{\pi^\star} - V_{G_\star}^{\widehat{\pi}} &= \left(V_{G_\star}^{\pi^\star} - V_{\widehat{G}}^{\pi^\star}\right) + \left(V_{\widehat{G}}^{\pi^\star} - V_{\widehat{G}}^{\widehat{\pi}}\right) + \left(V_{\widehat{G}}^{\widehat{\pi}} - V_{G_\star}^{\widehat{\pi}}\right) \\ &\le B + 0 + B. \end{aligned}$$

The middle term is nonpositive by optimality of $\widehat{\pi}$ in $\widehat{G}$. □

For $L_G \le 1$, this has the desired $O(\epsilon/(1-\gamma)^2)$ dependence. These are deterministic uniform-error bounds; marginal conformal coverage alone does not imply their assumptions. Related Lipschitz model-based bounds are developed in [1].

Logical separation. The two conformal propositions above are finite-sample distributional statements for a random transition or a predeclared behavior-policy rollout. The Lipschitz assumption in this section is instead deterministic and uniform over a forward-invariant state–action region and the policy comparison class. Neither type of statement implies the other. In particular, we do not replace the uniform ϵ in the policy-transfer bound by a marginal conformal radius; the scaling experiment tests the deterministic theorem, while the coverage experiments test the separate conformal statements.

The reproducible sharpness construction uses $G(x) = L_G x$, $\widehat{G}(x) = L_G x + \epsilon$, and the Lipschitz reward $-L_r|x|$. Across six values each of ϵ and γ, the measured log–log slope is 1.000 in ϵ. After dividing the value gap by $\gamma\epsilon$, its slope against the effective horizon $(1-\gamma)^{-1}$ is 2.000 up to numerical precision, exactly matching the closed form.

6 Experimental design

System. We use the controlled viscous Burgers equation

$$u_t + u u_x = \nu u_{xx} + g_{\mathrm{act}} a_t b(x),$$

with nonhomogeneous Dirichlet boundaries. Training and deployment differ in viscosity, boundary values, initial-condition amplitude, and the latent actuator gain g_{act}. The controller observes the state and boundary condition but not g_{act}.

Two-dimensional uncertainty benchmark. The official NeuralOperator Navier–Stokes benchmark [9] contains 128×128 input and output vorticity fields. We train a clean-label FNO and a perturbed-label FNO, then evaluate max-score simultaneous coverage on an independent split. Because this public dataset contains no action variable, it tests function-valued uncertainty calibration only and cannot support a closed-loop control claim.

Baselines. All methods share one trained residual FNO: (i) uncontrolled, (ii) nominal MPC, (iii) an L2 tube calibrated on source data, (iv) an L2 tube calibrated on the same deployment audit, (v) the audit-calibrated ellipsoid with adjoint support, and (vi) the same ellipsoid with adversarial nonlinear rollouts. The comparison between (iv)–(vi) isolates ambiguity geometry and inner-solver effects from access to deployment data.

Metrics. We report one-step set coverage, radius–error association, mean and median closed-loop cost, 90th-percentile and worst-case cost, paired seed-level differences, action magnitude, and max-over-time-and-coordinate trajectory coverage under the held-out behavior policy. A positive result requires that the proposed method improve at least a predeclared tail-risk metric without a material increase in mean cost. Model RMSE alone cannot establish the claim.

6.1 Preliminary diagnostic: why coverage is insufficient

Before introducing the conformal ellipsoid in Equation (1), we tested scalar L2 and fixed spatially weighted radii on three shifted rollouts. Deployment audit calibration increased combined-shift one-step coverage from 0.611 to 0.822–0.856, but no robust controller improved on the uncontrolled cost. This small smoke test is not a final comparison. It serves as a falsifying diagnostic: restoring predictive coverage did not by itself produce a useful decision. Quantitative claims remain preliminary until the paired study includes substantially more initial states.

6.2 Expanded controlled-Burgers evaluation

We trained the two residual FNOs for 60 epochs and evaluated 24 matched closed-loop cases of length 20 under the combined parameter, boundary, initial-condition, and hidden-actuator shift. The final perturbed-FNO validation MSE was 9.87×10^{-5}. Tables 1 and 2 separate the predictive and decision-facing conclusions. Source-calibrated L2 coverage fell to 0.741 under combined shift. Deployment-audit calibration restored the ellipsoid and simultaneous box to 0.906 function-space coverage. The box has the strongest projected-decision coverage, but also the largest mean support; coverage is therefore not interpreted as tightness.

Table 1: Combined-shift predictive results for the full seed-27 run. Decision coverage tests the residual projected onto the stage-cost direction; support is the corresponding ambiguity-set half-width.

Set	Function coverage	Decision coverage	Mean support
ID L2	0.741	0.820	0.00650
Audit L2	0.884	0.934	0.01072
Audit ellipsoid	0.906	0.986	0.02012
Audit simultaneous box	0.906	1.000	0.04422

Relative to nominal MPC, the simultaneous-box adjoint controller reduced mean cost by 1.89% and empirical $p90$ cost by 10.84%; the audit-L2 controller reduced them by 1.92% and 7.79%, respectively. The ellipsoidal adjoint and nonlinear adversarial controllers were less effective in this run. Thus the result supports a geometry-dependent control benefit, not universal dominance of a particular robust solver.

Table 2: Closed-loop cost on 24 matched combined-shift cases. Percentages are relative to nominal MPC; lower is better.

Controller	Mean	Mean change	$p90$	$p90$ change
Nominal MPC	0.5797	–	1.0752	–
ID-L2 robust MPC	0.5724	−1.27%	1.0250	−4.66%
Audit-L2 robust MPC	0.5686	−1.92%	0.9914	−7.79%
Ellipsoid-adjoint MPC	0.5747	−0.88%	1.0201	−5.12%
Box-adjoint MPC	0.5688	−1.89%	0.9586	−10.84%
Adversarial MPC	0.5764	−0.58%	1.0532	−2.04%

The separately calibrated max-over-time-and-coordinate band attained 0.88 empirical coverage on held-out length-10 behavior-policy trajectories, versus a nominal target of 0.90. This finite test proportion is reported rather than rounded into a guarantee. More importantly, the trajectory statement applies only to exchangeable behavior-policy rollouts and does not certify the counterfactual action sequences searched by MPC.

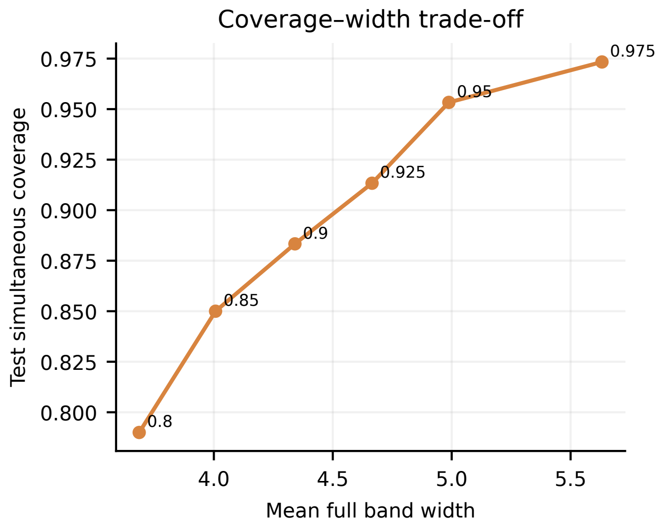

Figure 3: Simultaneous coverage on the independent NS2D test split increases only by accepting wider coordinate bands. This is a function-space uncertainty stress test, not a controlled-system result.

6.3 Official two-dimensional Navier–Stokes audit

On the official 128×128 benchmark we use 800 proper-training, 200 audit, and 300 independent test pairs. The max-normalized conformal score gives $q_{0.90} = 30.679$. Its empirical simultaneous test coverage is 0.883, while the mean one-step relative L2 error is 0.1635 and the mean full coordinate-band width is 4.3409. The test proportion is not rounded to the nominal 0.90. More importantly, the large max-score multiplier and band width expose a failure mode: simultaneous coverage over 128^2 output coordinates can be too conservative to define a useful planning set. The pointwise disagreement scale also correlates only $r = 0.169$ with absolute error, exposing weak spatial error localization. Because these pairs have no action channel, this experiment is evidence about function-space calibration and dimensionality, not robust control.

Table 3: Independent-test function-space uncertainty on the official NeuralOperator Navier–Stokes benchmark. Band width is the mean full coordinate-wise width.

Relative L2 error	Max-score $q_{0.90}$	Simultaneous coverage	Band width
0.1635	30.679	0.883	4.3409

6.4 Preliminary paired gain sweep

We next fixed the initial field, viscosity, and boundary conditions within each seed and evaluated all controllers at latent actuator gains $\{-0.5, 0, 0.5, 1, 1.5\}$. This removes the confounding between shift severity and initial-condition difficulty. Table 4 reports three initial seeds. The audit contains 90 independent deployment transitions.

Table 4: Preliminary paired-factorial evidence. Changes are relative to nominal MPC within each seed. Empirical $p90$ uses only five gain levels.

Seed	ID coverage	Audit coverage	Mean change	$p90$ change
0	0.283	0.922	+0.41%	−3.68%
1	0.206	0.950	−0.41%	−1.59%
2	0.222	0.878	+0.15%	−1.57%
Mean relative change	–	–	+0.05%	−2.28%

The adversarial controller improves the empirical upper-tail metric in all three seeds with negligible average change in mean cost. On the same calculation, the audit-calibrated L2 tube improves $p90$ by 1.74%. This supports the mechanism but not a population claim: each seed contains only five paired gains, and the nonlinear inner maximization is approximate.

6.5 Independent value-bound comparison

We compare four discounted-value bounds on disjoint calibration and test trajectories: a global maximum-error recursion, a state-dependent local recursion, adjoint support of the anisotropic set, and adjoint support plus an audit-calibrated quadratic feature. All four raw constructions are finally scaled on 60 calibration trajectories to the same 90% value-level target before evaluation on 160 independent test trajectories, with horizon 20 and $\gamma = 0.95$. The local recursion has the smallest mean bound, 0.0870, at 0.944 empirical coverage. Adjoint support is not tighter in this experiment: its mean bound is 0.2035 at 0.925 coverage. The audit-fitted curvature coefficient is zero, making the last two mean bounds identical; no curvature benefit is claimed. This negative result is important: decision weighting can improve a controller without automatically yielding a smaller scalar value certificate.

Table 5: Independent-test comparison at a matched 90% calibration target. Utilization is observed value gap divided by the reported bound.

Method	Coverage	Mean bound	Median η	p90 η	Max η
Global maximum	0.938	0.0987	0.210	0.688	3.919
Local recursion	0.944	0.0870	0.322	0.788	2.190
Adjoint support	0.925	0.2035	0.265	0.854	1.766
Adjoint + curvature	0.925	0.2035	0.265	0.854	1.766

7 Limitations and planned stress tests

Split conformal validity requires exchangeability between the audit and target deployment examples. One-step calibration gives marginal transition coverage; the separate max-score proposition gives simultaneous coverage only for a predeclared, exchangeable behavior-policy rollout. Neither statement certifies counterfactual action sequences chosen by MPC, nor does either imply the deterministic uniform-error assumption used by the value theorem. The robust objective is first order unless the curvature buffer is certified. The policy-transfer bound also assumes an exact learned-model optimizer, whereas the reported MPC uses a finite CEM search and is not certified by that theorem. The pointwise scale can be misspecified; a deep FNO ensemble and a learned uncertainty head are therefore required baselines [8]. Finally, one-dimensional Burgers control is a mechanism test, not evidence of broad PDE-control safety, while public uncontrolled Navier–Stokes pairs do not close that gap. Planned stress tests include audit-size sweeps, shift-severity sweeps, adversarial actuator degradation, paired random seeds, and an action-conditioned two-dimensional system.

8 Conclusion

Predictive uncertainty becomes useful for control only after its geometry is connected to the downstream objective and verified in closed loop. The proposed route is deliberately simple: conformalize one anisotropic dynamics ellipsoid, then optimize control against adverse transitions in that same set. The adjoint support provides a fast, interpretable approximation; projected adversarial rollout retains the nonlinear cost. Whether either conversion improves decisions is an empirical question, and the evaluation is designed to reject the method when it improves coverage but not control.

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